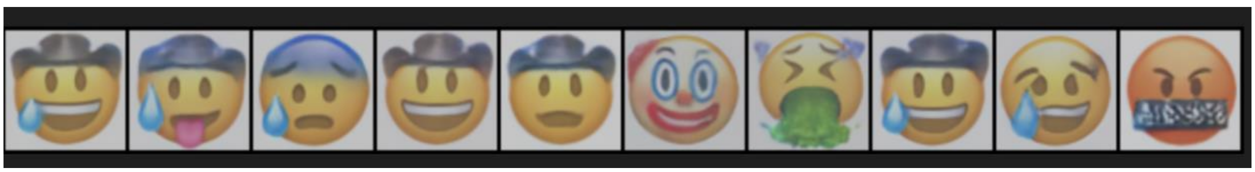
Homework 4

Section 2: Analysis of Mr. Potato Head like Generations

The generated samples show clear compositional behavior. The model does not simply copy one training face, but recombines semantically meaningful parts such as headwear, eyewear, mouth geometry, color tone, and tear patterns. Below are some novel examples (from different epochs) that exhibit this effect.

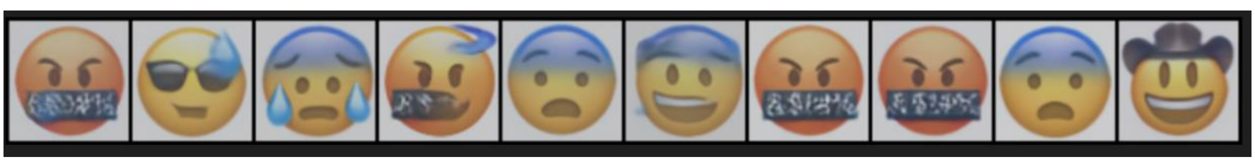
Epoch 600: This is a three-feature combination that merges a hat with a sweat drop and a stuck-out tongue.

- **Hat Source**: Row 5, Column E
- **Tear Source**: Row 7, Column A
- **Sweat Source**: Row 6, Column K and Row 6, Column L



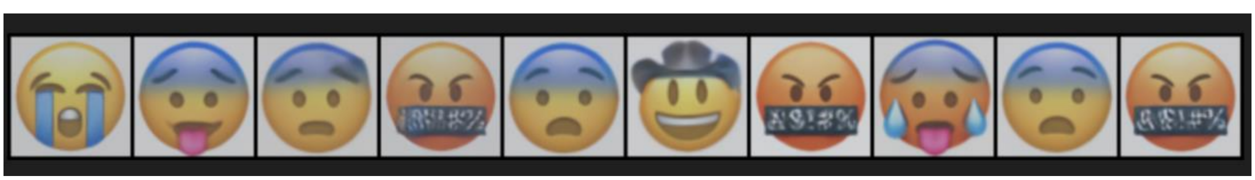
Epoch 850: This case combines glasses with a sweat drop located on the left side.

- **Specs Source**: Row 5, Column G
- **Sweat Source**: Row 4, Column K



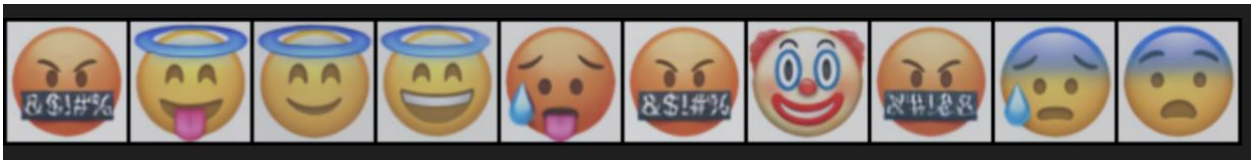
Epoch 1200: This is a three-feature combination utilizing an upper blue forehead, a red face tone, and tears on both sides of the face.

- **Cold-sweat Source**: Row 6, Column K
- **Red-face Source**: Row 5, Column B



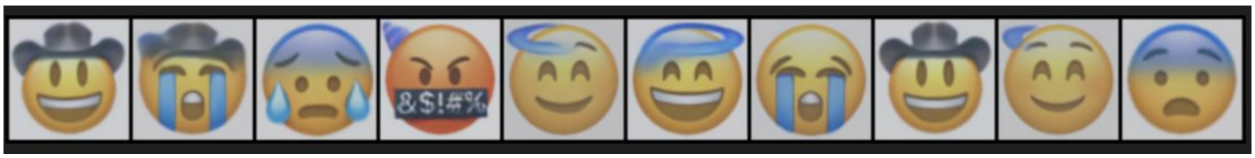
Epoch 1350: This generation combines a stuck-out tongue with a halo/angelic top feature.

- **Tongue Source**: Row 2, Column F
- **Halo Source**: Row 1, Column E



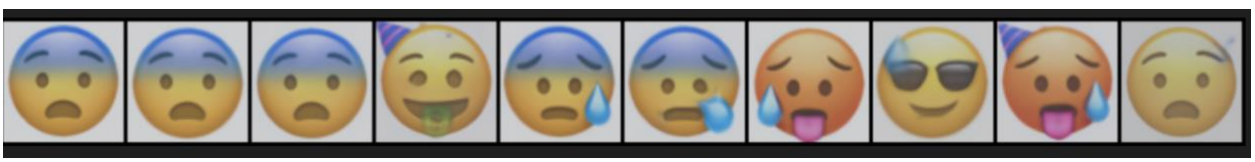
Epoch 1400: This sample merges a hat structure with heavy vertical tear streams.

- **Hat Source**: Row 5, Column E
- **Cry-stream Source**: Row 7, Column B



Epoch 1600: This is a three-feature combination, combining a birthday cap, a dollar-sign mouth, and neutral dot eyes.

- **Cap Source**: Row 5, Column F
- **Money-mouth Source**: Row 2, Column J
- **Eye-shape Source**: Row 3, Column E



Interpretation

Across epochs 600 to 1950, the model repeatedly combines identity-like traits (hat, glasses, cool-head styling) with emotion/expression traits (cry streams, sweat drop, tongue-out geometry, flushed tone). This repeated cross-feature recombination is consistent with the Mr. Potato Head effect discussed in class and supports the claim that the diffusion model has learned compositional structure instead of only nearest-neighbor memorization.